Infusion Pump
Operation Procedures

Project: Infusion pump

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1 Intended Operating Environment

The infusion pump is designed for use in professional healthcare facilities, including but not limited to:

- Hospitals
- Intensive Care Units (ICU)
- Operating Rooms (OR)
- Emergency Departments
- General wards

The device shall be operated only by trained and authorized medical personnel in accordance with hospital policies and clinical protocols.

1.1 Environmental Operating Conditions

1. Temperature
    - Operating temperature range: +10 °C to +40 °C
    - Storage temperature range: −20 °C to +60 °C
2. Relative Humidity
    - Operating humidity: 15% to 90% RH, non-condensing
    - Storage humidity: 10% to 95% RH, non-condensing
3. Atmospheric Pressure
    - Operating pressure range: 700 hPa to 1060 hPa
4. Ingress Protection
    - The device shall meet a minimum ingress protection rating of IPX1 (protection against vertically falling water drops), or higher as required by clinical use.
5. Vibration and Shock
    - The device shall operate safely under normal hospital transport conditions, including movement on carts and bedside mounting.

1.2 Electrical and Power Conditions

1. The infusion pump shall operate from:
    - Internal rechargeable battery, and/or
    - External AC power source compliant with IEC 60601-1
2. The device shall tolerate temporary power interruptions without loss of critical infusion data or unsafe interruption of therapy.
3. Battery operation shall support continued infusion for a minimum specified duration under normal operating conditions.

1.3 Electromagnetic Compatibility (EMC)

1. The device shall comply with IEC 60601-1-2 requirements for electromagnetic compatibility.
2. The infusion pump shall operate safely in the presence of common hospital electromagnetic sources, including:
   - Other medical devices
   - Wireless communication equipment
   - Mobile computing devices

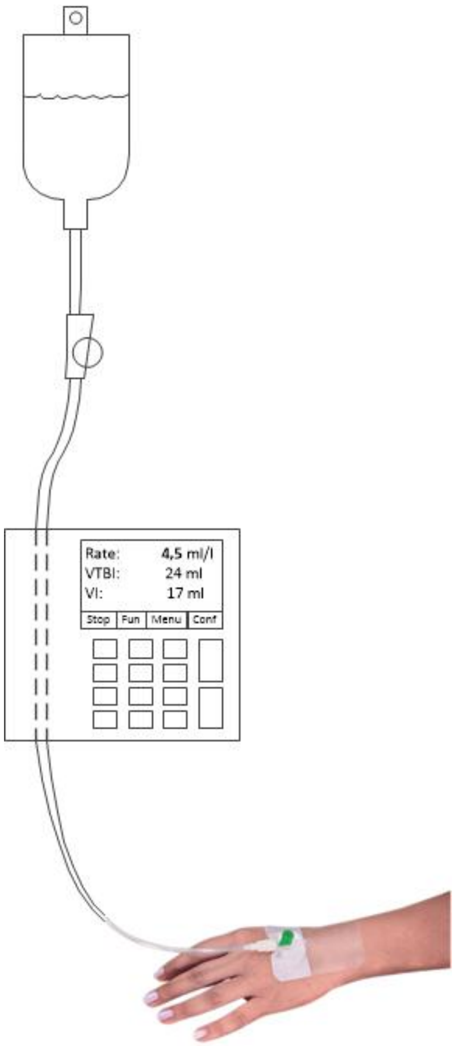
1.4 Wi-Fi Network Availability and Communication Requirements

1. Network Availability
   - The infusion pump shall be capable of operating safely and independently of Wi-Fi network availability.
     - Infusion delivery shall not depend on continuous network connectivity.
     - Loss of Wi-Fi connectivity shall not interrupt active infusions.
   - The device shall store critical operational and therapy data locally during network outages and synchronize automatically when connectivity is restored.
2. Supported Network Characteristics
   - The Wi-Fi module shall support:
     - IEEE 802.11 a/b/g/n/ac (as applicable to hospital infrastructure)
     - Operation on 2.4 GHz and/or 5 GHz bands
   - The device shall support enterprise-grade hospital network security, including:
     - WPA2-Enterprise and/or WPA3-Enterprise
     - Authentication via hospital-approved methods (e.g., certificates, EAP-TLS)
3. Network Performance Requirements
   - The device shall maintain reliable communication with the hospital server under the following minimum network conditions:
     - Packet loss: ≤ 5%
     - Latency: ≤ 500 ms for non-critical data exchange
     - Bandwidth sufficient for periodic data uploads and command reception
   - The device shall automatically attempt reconnection in the event of temporary network loss without requiring user intervention.

2 Intravenous infusion equipment

Checklist for infusion equipment.

1. …
2. …
3. …
4. …
5. …



3 Intravenous infusion procedure

3.1 Infusion preparation

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2. …
3. …
4. …
5. …

3.2 Infusion supervision

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2. …
3. …
4. …
5. …

3.3 Infusion medical documentation

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2. …
3. …
4. …
5. …